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Anillo polinomios

Proposición 1 (Anillo de polinomios). *Sea A un anillo (conmutativo con unidad)*

$$\implies A[x] := \{a_n x^n + \cdots + a_1 x + a_0 : a_i \in A \wedge n \in \mathbb{N}\}$$

es un anillo (conmutativo con unidad) con las siguientes operaciones:

- *Suma:* $(a_n x^n + \cdots + a_0) + (b_m x^m + \cdots + b_0) = (a_n + b_n) x^n + \cdots + (a_0 + b_0)$.
- *Producto:* $(a_n x^n + \cdots + a_0) \cdot (b_m x^m + \cdots + b_0) = \displaystyle\sum_{k=0}^{n+m} \left(\sum_{i=0}^{k} a_i b_{k-i} \right) x^k$.

Lo llamamos el anillo de polinomios en la variable x con coeficientes en A.

Referenciado en

- `ralgebra-finitamente-generada`
- `elemento-entero-sobre-anillo`
- `lem-anillo-polinomios-variables-cuerpo-ideal-maximal-extension-algebraica-grado-fin`
- `teo-ideal-maximal-anillo-polinomios-cuerpo-alg-cerrado`
- `prop-grado-polinomio`
- `funcion-regular-variedad-algebraica-afin`
- `elemento-algebraico-sobre-anillo`
- `ideal-anulacion`
- `lem-con-ceros-ideal-generado`
- `teo-con-ceros-polinomios-esp-afin-ideal-propio-cuerpo-alg-cerrado-no-vacio`
- `teo-base-hilbert`
- `alg-cerrado`
- `independencia-algebraica`

- obs-con-ceros-ideal-generado-contenido-trivial
- grado-polinomio
- polinomio-monico-variable
- con-ceros-polinomios-esp-afin
- variedad-algebraica-afin
- ejer-variedad-algebraica-ideal-radical
- cor-base-hilbert-n-var
- anillo-coordenadas-variedad-algebraica-afin
- teo-cuerpo-imp-polinomios-dip
- prop-variedad-algebraica-afin-ideal
- teo-ceros-hilbert
- prop-di-imp-anillo-polinomios-di

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